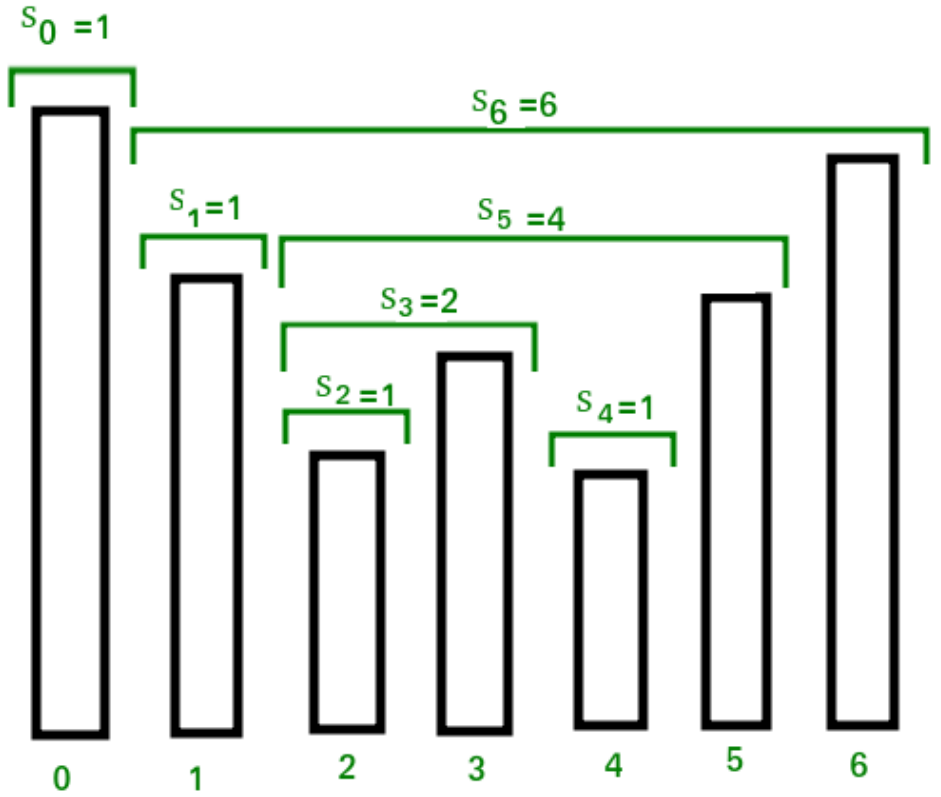
DS	**Session**	Stack	**Question Information**	Level 1 Challenge 46

Question description

The stock span problem is a financial problem where we have a series of n daily price quotes for a stock and we need to calculate span of stock's price for all n days.
The span Si of the stock's price on a given day i is defined as the maximum number of consecutive days just before the given day, for which the price of the stock on the current day is less than or equal to its price on the given day.
For example, if an array of 7 days prices is given as {100, 80, 60, 70, 60, 75, 85},

then the span values for corresponding 7 days are {1, 1, 1, 2, 1, 4, 6}



Constraints

0<n<1000

0<price[i]<1000000

Input Format:

First line indicates the number of days

second line indicates the price quoted for above mentioned days

Output Format:

Single line represents the span values for corresponding days

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6
9 8 7 6 5 4
```

EXPECTED OUTPUT

```
1 1 1 1 1 1
```

Test Case 2

INPUT (STDIN)

```
6
10 41 15 90 10 70
```

EXPECTED OUTPUT

```
1 2 1 4 1 2
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void printArray(int
arr[],int n)
```

Test Case 2

KEYWORD

```
void calculateSpan(int
price[],int n,int S[])
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
4
```

Test Case 2

TOKEN COUNT

```
220
```

Test Case 3

NLOC

```
35
```

Code

solution.c

```c
#include <stdio.h>

void calculateSpan(int price[], int n, int s[]) {
    int st[1000], top = -1;
    st[++top] = 0;
    s[0] = 1;
    for (int i = 1; i < n; i++) {
        while (top >= 0 && price[st[top]] <= price[i]) top--;
        s[i] = (top == -1) ? (i + 1) : (i - st[top]);
        st[++top] = i;
    }
}

void printArray(int arr[], int n) {
    for (int i = 0; i < n; i++) {
        printf("%d%s", arr[i], (i == n - 1) ? "" : " ");
    }
    printf("\n");
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int price[n], s[n];
    for (int i = 0; i < n; i++) scanf("%d", &price[i]);
    calculateSpan(price, n, s);
    printArray(s, n);
    return 0;
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
6
9 8 7 6 5 4
1 1 1 1 1 1
PS C:\C Programming>
```